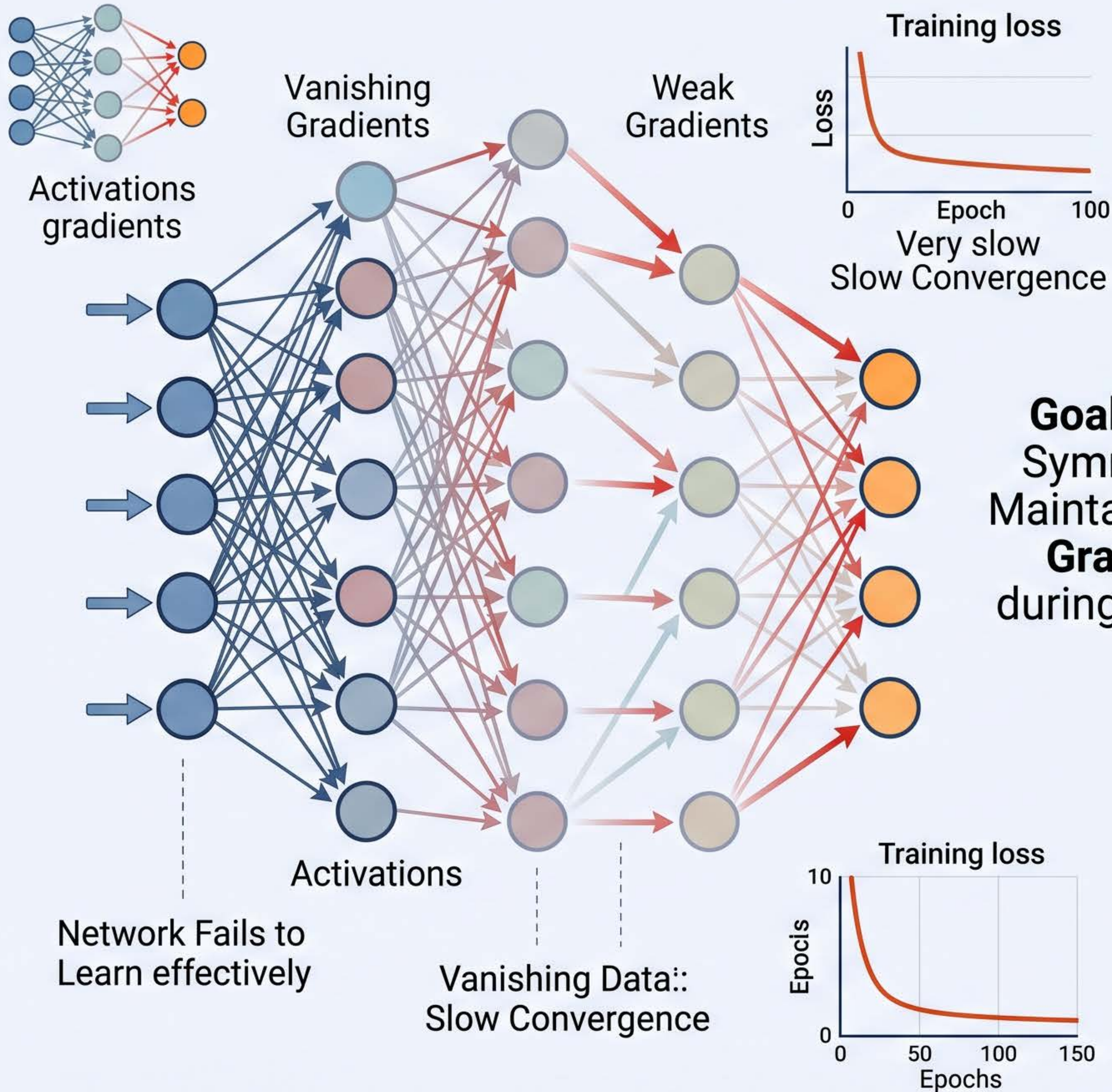


POOR INITIALIZATION

(e.g., Small Random or All Zeros)

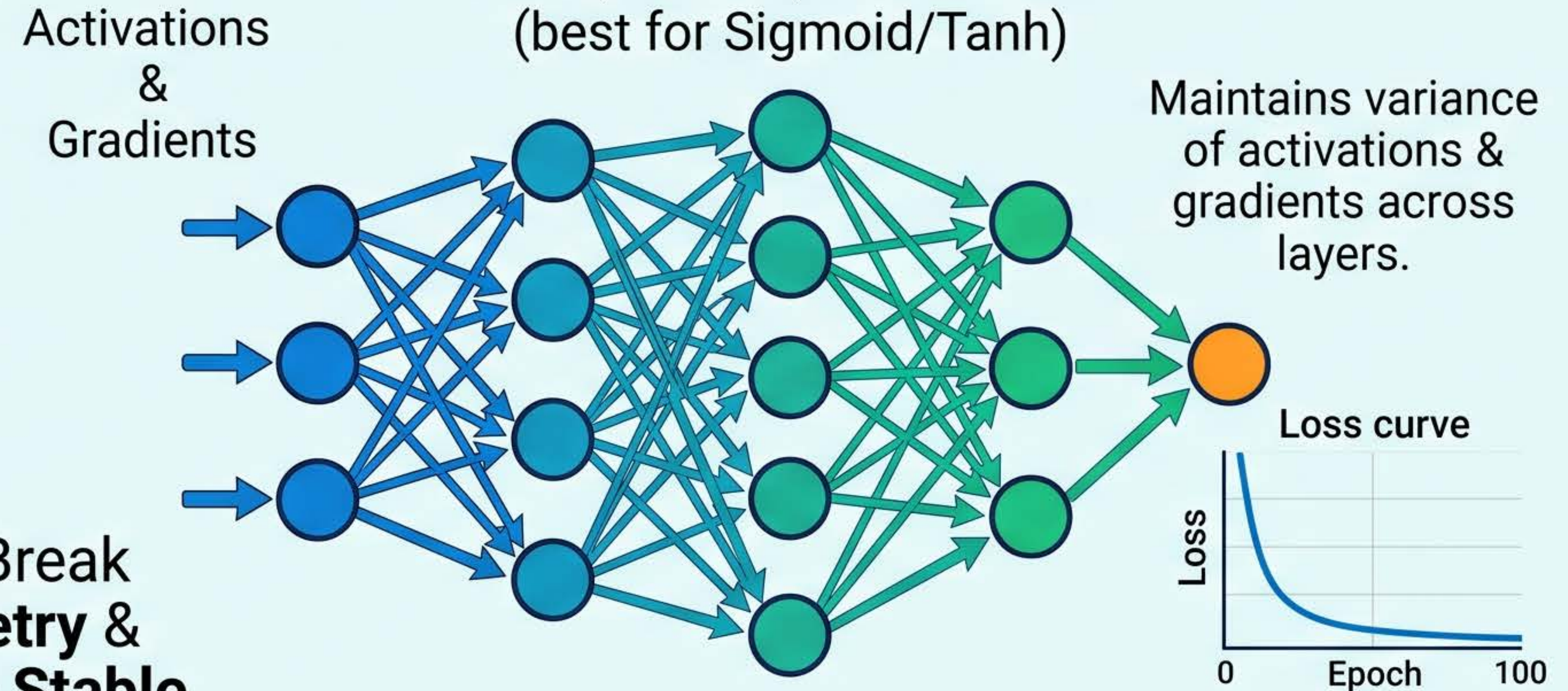


Goal: Break Symmetry & Maintain Stable Gradients during Training

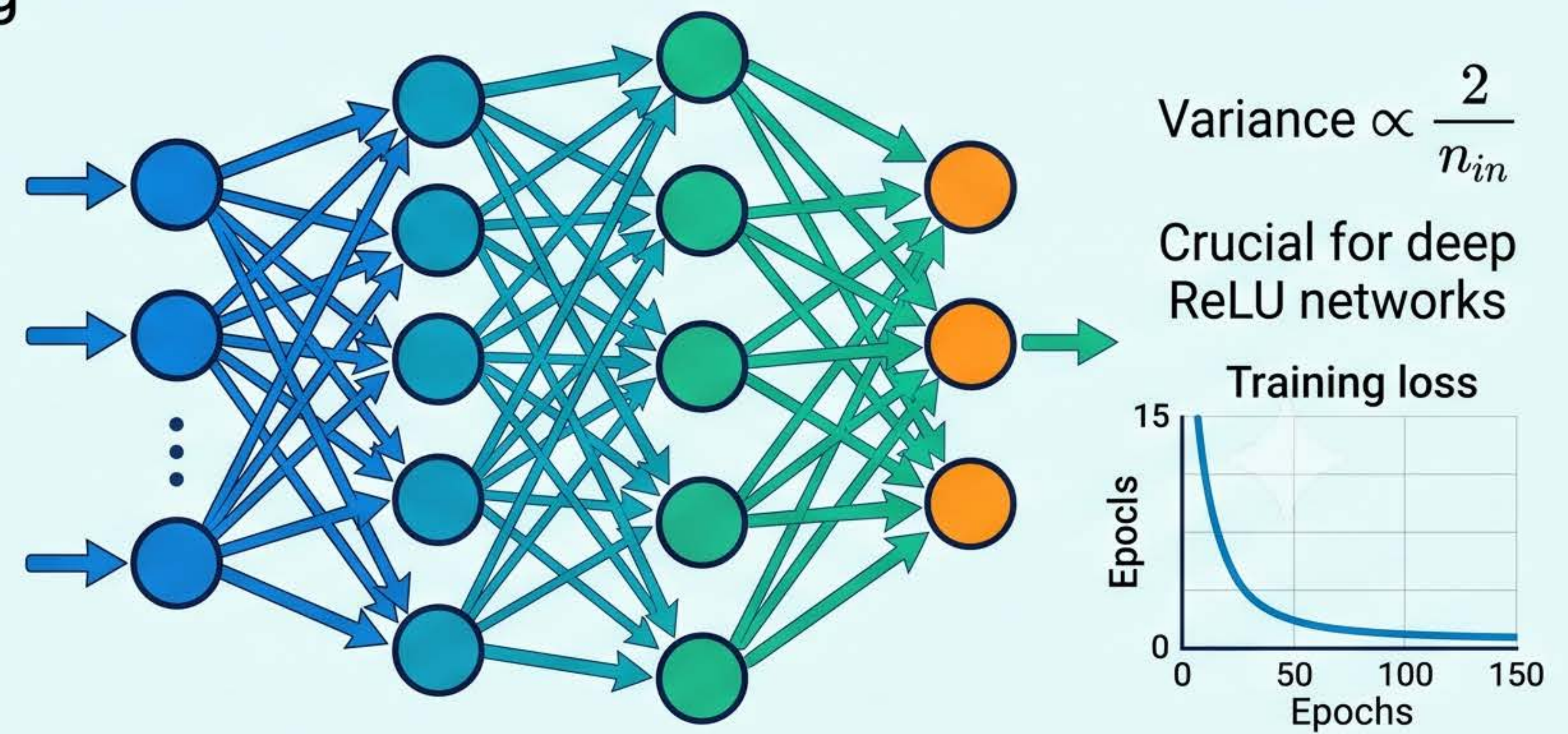
BETTER WEIGHT INITIALIZATION METHODS

Xavier (Glorot) Initialization

(best for Sigmoid/Tanh)



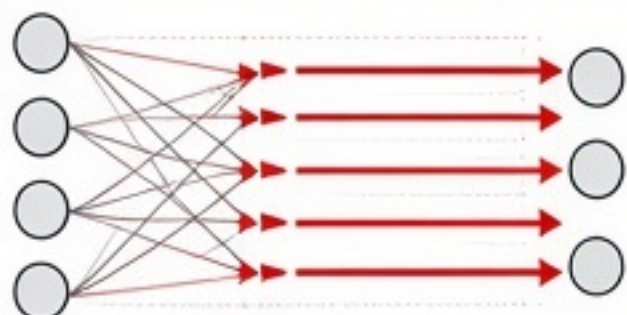
He Initialization (best for ReLU)



Better Weight Initialization Methods

for Deep Neural Networks

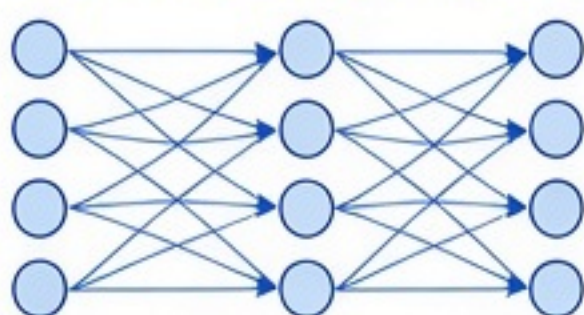
1. Naive Random Initialization



- ⚠ Exploding Gradients (too large)
- ⚠ Vanishing Gradients (too small)

Unstable training: gradients explode or vanish, causing poor convergence or divergence.

2. Xavier / Glorot Initialization



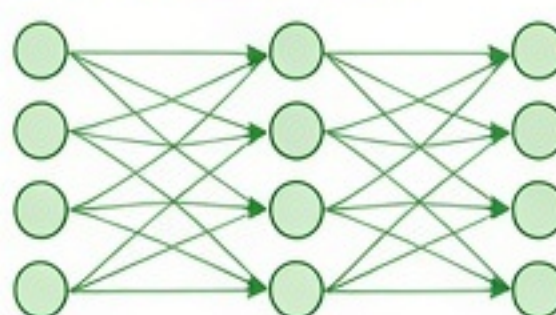
Weight Initialization:

$$W \sim \text{Uniform}\left(-\sqrt{\frac{6}{fan_{in} + fan_{out}}}, \sqrt{\frac{6}{fan_{in} + fan_{out}}}\right)$$

or $\text{Normal}\left(0, \sqrt{\frac{2}{fan_{in} + fan_{out}}}\right)$

- ✓ Balanced gradients, good for sigmoid / tanh activations.

3. He Initialization

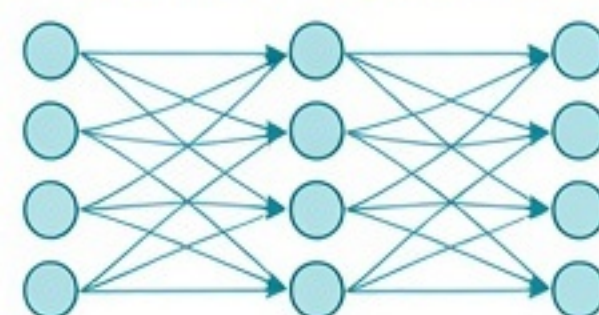


Weight Initialization:

$$W \sim \text{Normal}\left(0, \sqrt{\frac{2}{fan_{in}}}\right)$$

- ✓ Suitable for ReLU (and variants) healthy gradient flow.

4. Modern / Orthogonal Initialization



Weight Initialization:

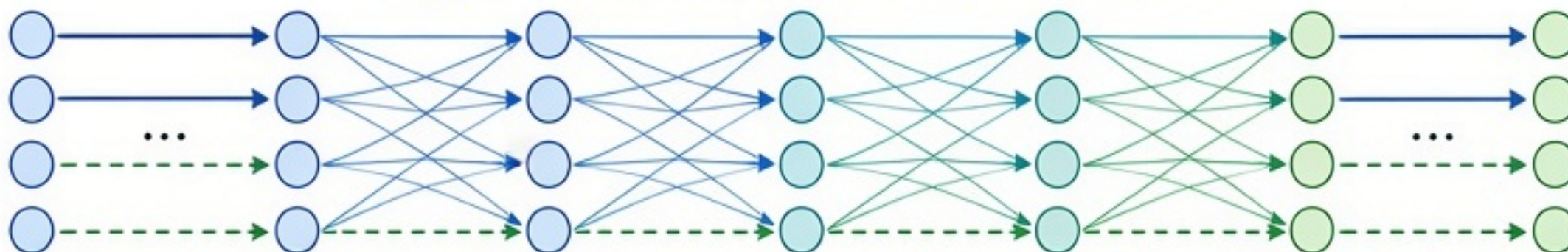
W orthogonal ($W^T W = I$)
or scaled variant
 $W = \sigma \cdot Q$ (Q orthogonal)

- ✓ Stable signal propagation across very deep networks.

Deep Neural Network with Proper Initialization

Forward Pass (Activations)

Proper variance of activations is preserved layer to layer.



Backward Pass (Gradients)

Gradients flow back without vanishing or exploding.



Proper Activation Variance
 $\text{Var}(a^{[l]}) \approx \text{Var}(a^{[l-1]})$
(stable forward signal)



Healthy Gradient Flow
Gradients have stable variance across layers.



Stable Training
Faster convergence, better performance, deeper networks trainable.